

Vikram Rangarajan

<https://vikramrangarajan.github.io/> | <https://github.com/VikramRangarajan/> | <https://linkedin.com/in/vikram-rangarajan/>
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Education

Purdue University, West Lafayette, IN, 47907

08/25 - Expected 05/30

Ph.D. -- Computer Science, AI / ML

GPA: 3.85

Advisor: Rajiv Khanna

Relevant Coursework: Foundations of Deep Learning, Randomized Algorithms, Statistical Machine Learning, Computer Networks

Topics of Interest: Deep Learning, Model Quantization, Second Order Methods, and GPU Kernels

University of Maryland, College Park, MD, 20742

09/22 - 05/25

B.S. -- Computer Science - Machine Learning Track

GPA: 4.0, Summa Cum Laude

Minor: Statistics

Relevant Coursework: Artificial Intelligence (AI), Machine Learning (ML), Computer Vision (CV), Natural Language Processing (NLP), Data Science (DS), Parallel Computing, Calculus 1, 2, & 3, Statistics, Linear Algebra, Compilers, Algorithms

Technical Skills

- **Programming Languages:** Python, Triton, C/C++/CUDA, Rust, Java, OCaml, R, Racket, Assembly (x86, MIPS), SAS
- **Machine Learning:** Hugging Face Ecosystem (transformers, datasets, trl, etc.), vLLM, PyTorch, JAX, NumPy, scikit-learn, OpenCV
- **HPC / Systems:** Linux, SLURM, Docker, Git, MPI, OpenMP, Ray
- **Data Engineering:** Azure Cloud Services, SQL, Relational Databases (Postgres, Oracle, SQL Server), Apache Airflow

Experience & Projects

Purdue University, West Lafayette, IN, 47907

08/25 - Present

Graduate Teaching Assistant

- Assisting with CS240: Programming in C
- Leading lab section to guide students to success
- Holding office hours to assist students with their coursework
- Grading assignments, proctoring exams, and assisting with course materials

IITM Pravartak Technologies Foundation, Chennai, Tamil Nadu, India, 600113

05/26 - 07/26

Machine Learning Intern

- Performed supervised fine-tuning and reinforcement learning (GRPO) to train a NL-to-cypher LLM
- Created a change detection training pipeline and added an IJEPA trainer to torchgeo
- Added dynamic KV-cache quantization to vLLM and Gemma 4 using posterior sampling and sharpness approximation

UMIACS, College Park, MD, 20742

10/24 - 06/25

Undergraduate Research Assistant

- Created a neural video codec to surpass state of the art compression algorithms for image and video data
- Models were fitted to decode the original video from input pixel coordinates efficiently
- Used methods such as model quantization and meta learning to achieve ideal reconstruction quality with high compression, high encoding, and high decoding speeds

Shahoveisi Lab, College Park, MD, 20742

02/24 - 11/24

Undergraduate Research Assistant

- Created manuscripts for machine learning research projects related to identifying and managing turfgrass related diseases
- Used methods such as transfer learning and gradual unfreezing to train highly accurate nematode image classifiers
- Performed automatic hyperparameter optimization using Ray Tune to train sklearn and torch models to achieve highest metrics
- Performed parallelized automatic image dataset preprocessing using OpenCV and NumPy

SimpleTensor

02/24 - 05/24

- Created a library which provides Tensors with reverse-mode automatic differentiation capabilities for the Intro to AI class
- Supports many differentiable n-dimensional tensor operations such as matmul, ConvNd, element-wise, reductions, etc.
- Created MNIST demo using convolutional, dense, and softmax layers
- Fully documented using sphinx at <https://vikramrangarajan.github.io/SimpleTensor/>

A.M. Best Rating Services, Oldwick, NJ, 08858

06/23 - 01/24

Data Strategy Engineer

- Gained advanced experience with relational databases, Docker, Linux, Python, and Pandas
- Learned to use Azure Data Factory (ADF) to transform and move data on the Azure Cloud Platform
- Used Apache Airflow to orchestrate ETL pipelines between on-prem databases and Azure
- Accelerated a data pipeline's execution time from 90 minutes down to 6 minutes using ADF

Publications

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|----|---|-----------------|
| 1. | Vikram Rangarajan , Fereshteh Shahoveisi, Benjamin Waldo, Sadegh Jafari
Identification of Plant-Parasitic Nematode Genera in Turfgrass Using Deep Learning Algorithms
Scientific Reports | Published 12/25 |
| 2. | Vikram Rangarajan , Shishira Maiya, Max Ehrlich, Abhinav Shrivastava
SIEDD (Shared Implicit-Encoder with Separate Decoders) | arXiv 06/25 |

Awards & Certifications

UMD CMNS Latin Honors - Summa Cum Laude	Spring 25
UMD Computer Science Semester Academic Honors	Fall 22 - Spring 25
Astronomer Certification for Apache Airflow Fundamentals	02/24